3D Creative Club Study Guide

Egg Model and Paint

Goal: What You Will Build

Object Mode



Steps:

- Model a simple egg shape from a UV Sphere
- Switch to Texture Paint mode
- Paint colors and patterns directly on the egg surface
- Create a bump map from your painted colors
- (Optional) Connect materials and add lighting
- After the tutorial, use your imagination to create something unique!

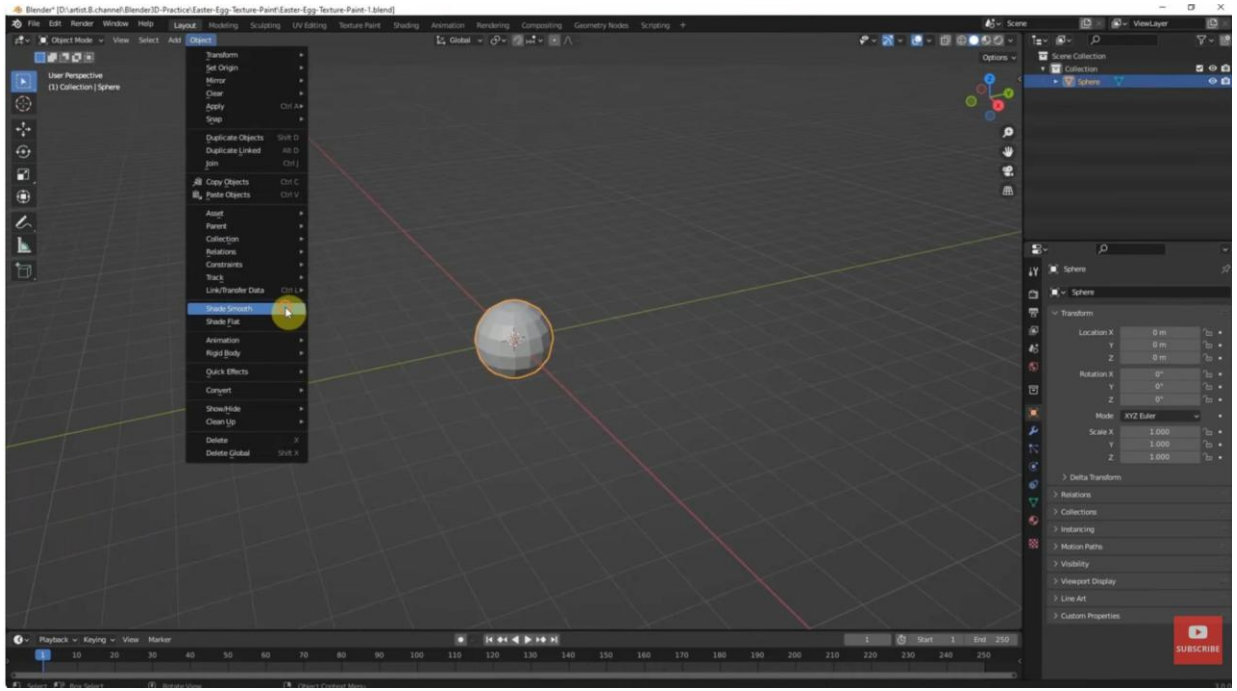
Shortcuts:

Shift + A: Add · Tab: Mode toggle · F: Brush size

Tip: This week you learn Texture Paint — painting directly on 3D objects. Steps 7-8 are optional.

Step 1: Add a Sphere & Smooth It

Object Mode



Steps:

- Press X to delete the default cube
- Shift + A → Mesh → UV Sphere
- Right-click the sphere and choose Shade Smooth
- Shade Smooth makes the surface look round instead of faceted

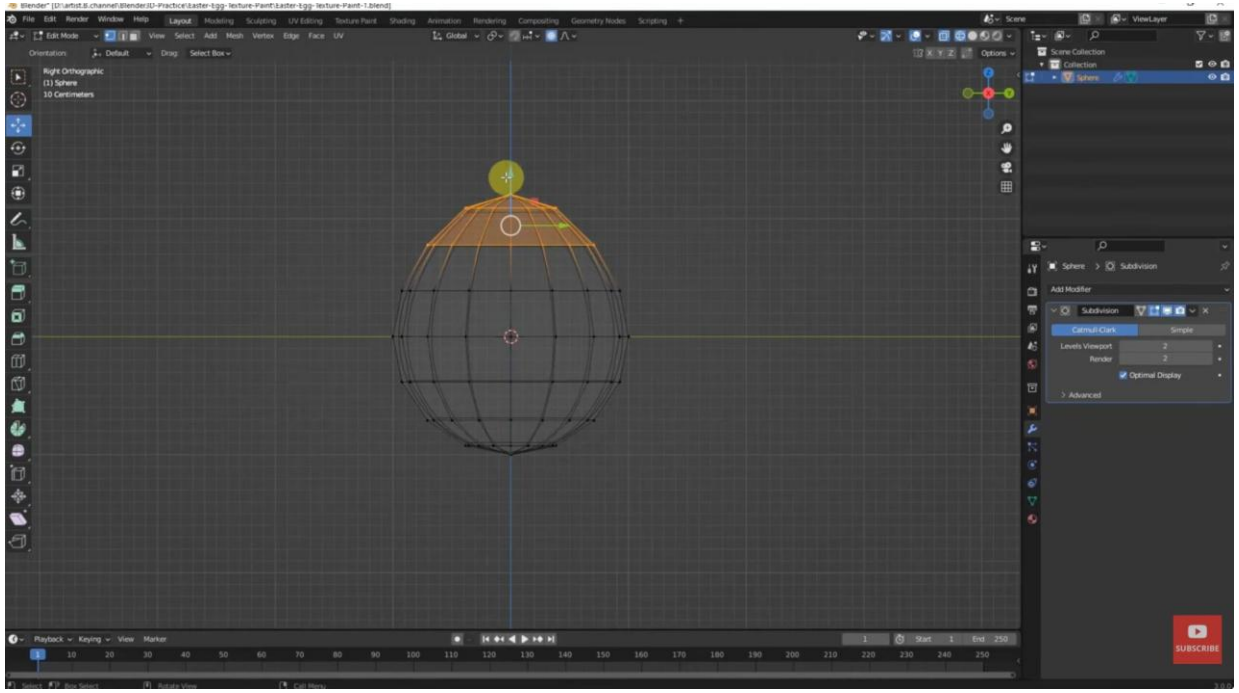
Shortcuts:

Shift + A: Add Sphere · Right click: Shade Smooth · X: Delete

Tip: A smooth sphere is easier to paint on. You can also add a Subdivision Surface modifier if needed.

Step 2: Shape the Egg

Edit Mode



Steps:

- Press Tab to enter Edit Mode
- Select the top vertices and scale down (S, then move mouse)
- Select the bottom vertices and scale up slightly
- The sphere should now look like an egg — narrower on top, wider on bottom

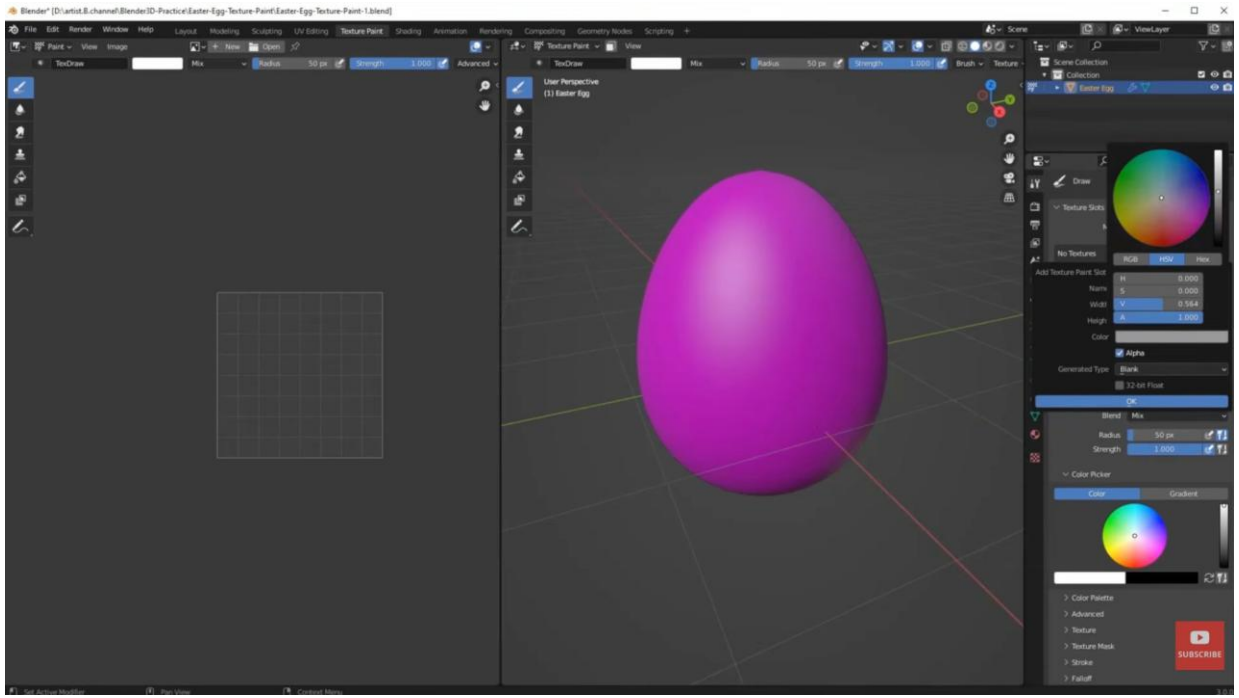
Shortcuts:

Tab: Edit Mode · S: Scale · G: Move · `: View selector

Tip: Use ` (backtick) to switch to Front view so you can see the profile clearly.

Step 3: Enter Texture Paint Mode

Texture Paint



Steps:

- Switch to Object Mode (Tab)
- Click the "Texture Paint" workspace tab at the top
- Blender will ask to create a new texture — click OK
- Set the image resolution to 1024 x 1024
- You now see the 3D egg on the left and a 2D paint canvas on the right

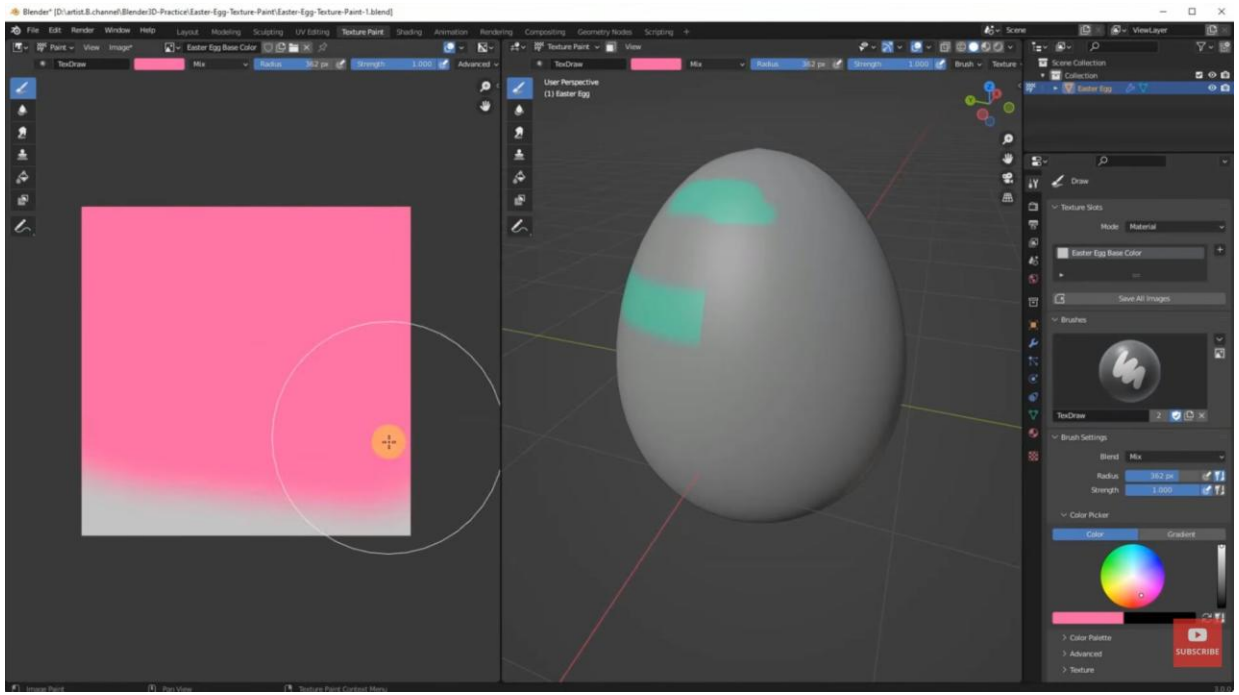
Shortcuts:

Tab: Object Mode · F: Brush size · Shift + F: Brush strength

Tip: The 2D canvas on the right shows the "unwrapped" texture map — painting here or on the 3D model both work.

Step 4: Paint Your Base Colors

Texture Paint



Steps:

- Pick a color from the color wheel on the right panel
- Choose a large brush size (press F and drag)
- Paint broad color bands or patches on the egg
- Try different colors for different sections
- The 2D canvas updates as you paint on the 3D model

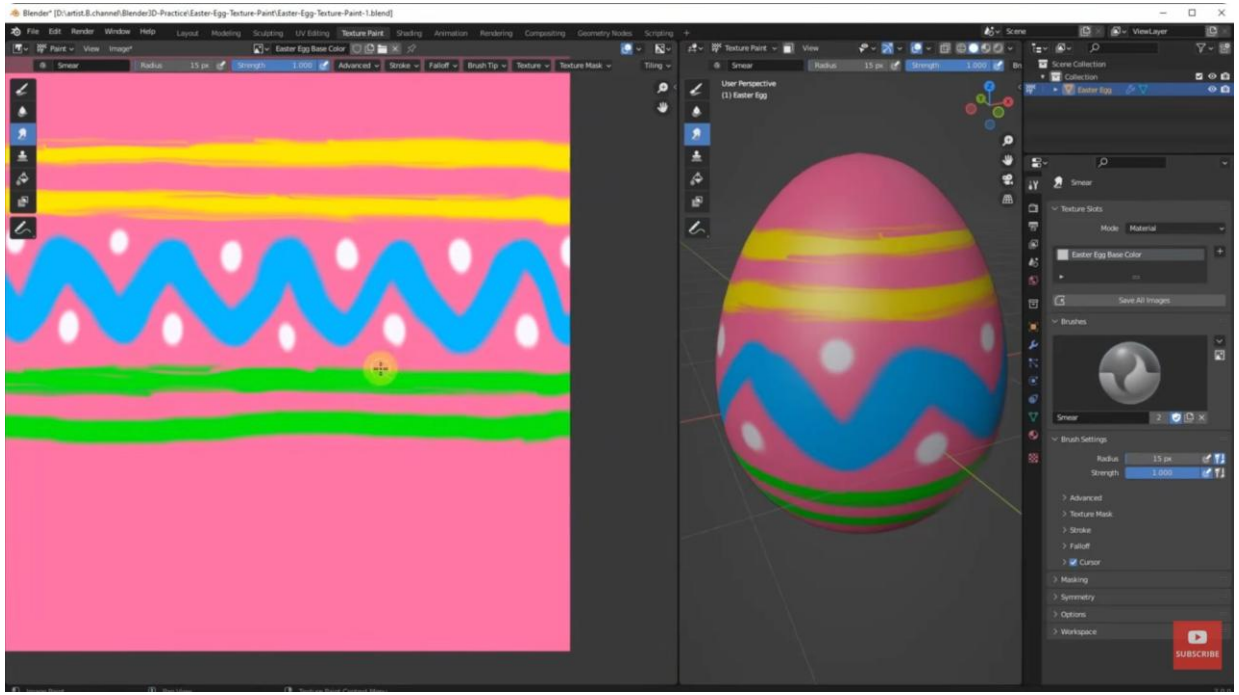
Shortcuts:

F: Brush size · Shift + F: Strength · Ctrl + F: Falloff

Tip: Start with big, simple shapes. You can add detail in the next step.

Step 5: Smear & Blend

Texture Paint



Steps:

- Switch the brush type to Smear
- Drag across color edges to blend them together
- Use this to create smooth gradients and wavy patterns
- Add extra details like dots or highlights

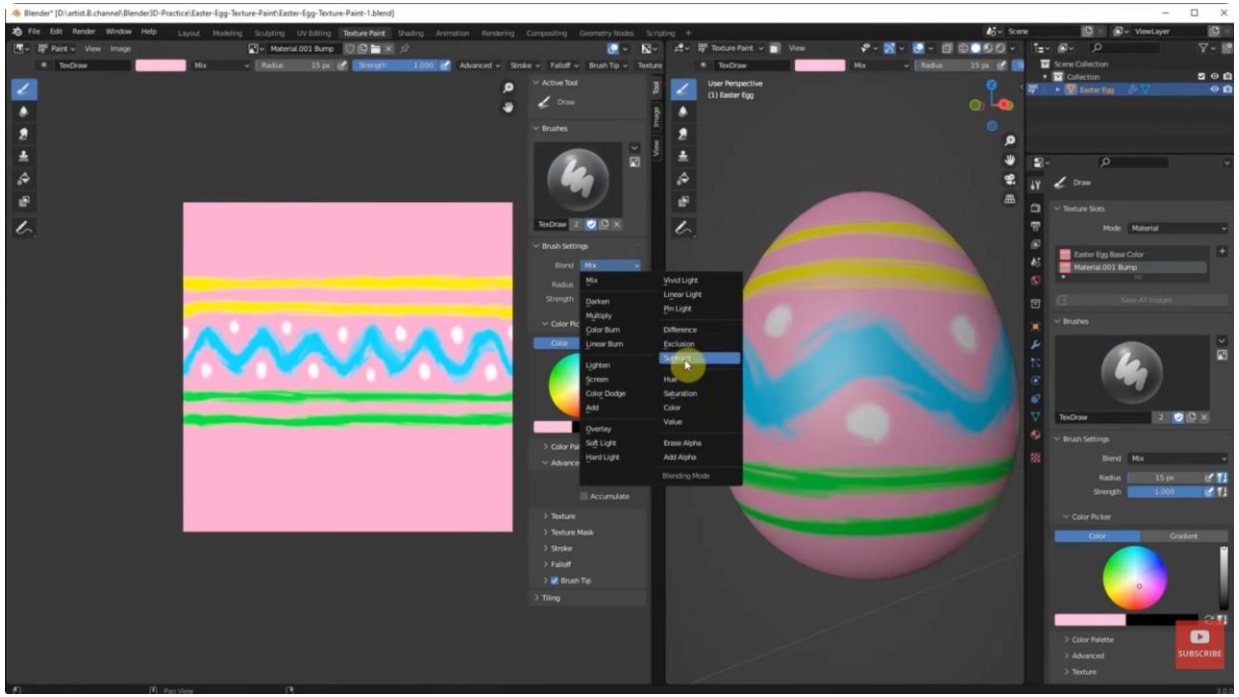
Shortcuts:

F: Brush size · Shift + F: Strength · Brush menu: Smear

Tip: The Smear brush is key to this tutorial's look — it blends painted colors into smooth gradients.

Step 6: Create a Bump Map

Texture Paint



Steps:

- Open the Image Editor and view your painted texture
- Create a greyscale (black & white) version of the texture
- This greyscale image becomes your bump / height map
- White areas = raised, dark areas = lower
- The bump map adds surface detail without extra geometry

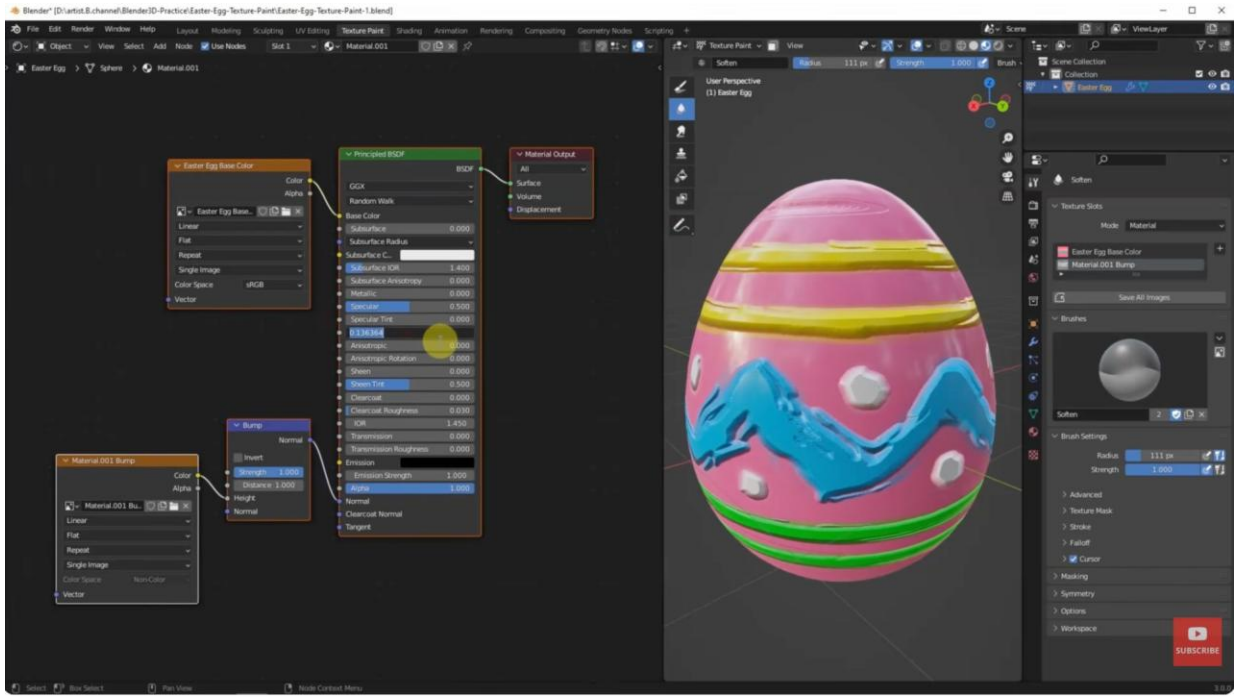
Shortcuts:

Ctrl + Tab: Mode switch · `: View selector

Tip: Bump maps fake surface detail using shading — the egg looks textured even though it is still smooth.

Step 7: Connect Materials (Optional)

Shading



Steps:

- Switch to the "Shading" workspace tab
- Select the egg, then click "New" in Material Properties
- Connect your color texture to the "Base Color" input
- Connect your bump texture to "Bump" → "Height" input
- Adjust Roughness and Specular to fine-tune the look

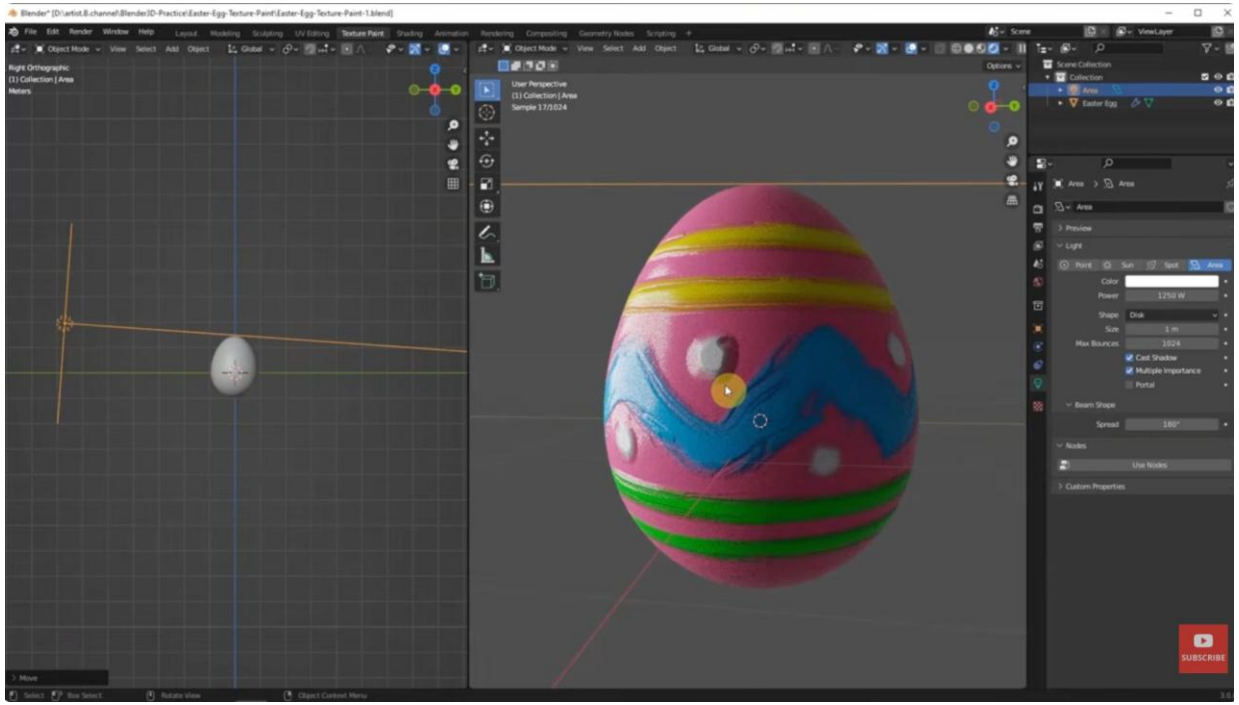
Shortcuts:

Ctrl + Tab: Mode · Z: Toggle shaded · Shift + click: Multi-select

Tip: This step is optional. The tutorial uses the Principled BSDF node to make the final render look polished.

Step 8: Set Up Lighting & Scene (Optional)

Object Mode



Steps:

- Add a plane (Shift + A → Mesh → Plane) as a backdrop/floor
- Scale the plane large enough to be a background
- Add a light: Shift + A → Light → Area Light (or Point Light)
- Position the light above and to the side of the egg
- (Optional) Press F12 to render the final image

Shortcuts:

Shift + A: Add Light/Camera/Plane · G/S/R: Transform · F12: Render

Tip: This step is optional. Good lighting makes the texture detail visible, but it is not required to finish the egg.

Creative Challenge: Make It Your Own!

Creative



What to do:

• You followed the tutorial — now use your imagination!

• Apply texture paint to ANY object you want:

• → Paint a dragon egg with scales and fire colors

• → Texture a rocket ship with sci-fi panels

• → Decorate a planet with oceans and continents

• → Design a game weapon or item with worn paint

• → Create a holiday ornament or gift box

• Use everything you learned: modeling, texture paint, bump maps, materials, and lighting!

• Upload your render to Teams for peer review.

• Most likes wins a 3D-printed prize next class!

Tip: Be original — the best work comes from your own ideas, not just copying the tutorial.

Shortcut Cheat Sheet

Shift + A	Add object (Sphere, etc.)	**Tab**	Toggle Edit / Object Mode	**S + X/Y/Z**	Scale on one axis
G + X/Y/Z	Move on one axis	**Right click**	Shade Smooth	**F**	Brush size (Paint mode)
Shift + F	Brush strength	**Ctrl + F**	Brush falloff	**Ctrl + Tab**	Mode pie menu
` (Backtick)	View selector pie menu	**Z**	Toggle Shaded / Wireframe	**F12**	Render image